

Akshay Singla

✉ akshaysingla0010@gmail.com | 📞 +91-9464892690 | 🔗 [linkedin.com/in/akshaysingla90](https://www.linkedin.com/in/akshaysingla90) |
🐙 github.com/akshaysingla90

Summary

6+ years of experience building and **scaling distributed systems and microservice architectures** in production, leveraging asynchronous programming, event-driven workflows, and batch processing with Node.js, NestJS, TypeScript, MongoDB, Redis, and Kafka. Specialized in Payroll, HRMS, and ERP domains, owning services processing payroll for 200k+ employees and delivering 200× latency improvements and sub-5-minute payroll execution at scale

Skills

Languages: JavaScript, TypeScript, Java, SQL

Frameworks: Node.js, Express.js, NestJS, React.js, Socket.io, Mocha

Databases: MySQL, PostgreSQL, MongoDB

AI Systems: RAG, Vector DBs(pgvector), LLM Workflows, MCP servers

Others: Kafka, Redis, Jenkins, Kubernetes, Docker, Git, AWS(EC2, S3, Lambda, Load Balancer)

Experience

Betterplace — HRMS Platform (30M+ Frontline Workers)

Bangalore, India

SDE 3

April 2025 - Present

- Designed and optimized distributed database architectures and **40+ NestJS APIs** in different feature modules of **HRMS payroll product** to process over **1.5B+ USD** in annual payroll volume for **1,500+ enterprises** with zero data loss.
- Architected and shipped a standalone **Doc Generator Service (NestJS, Puppeteer, Kafka, AWS S3)** extracted from the payroll monolith to offload payslip and income-tax computation rendering — sustained generation for **10k+ employees concurrently** from scratch without degrading the payroll critical path.
- Redesigned the payroll job-processing pipeline using **BullMQ + Redis** implementing **batch processing** with independent concurrent workers; cut end-to-end runtime for **200k-employee** payrolls from **hours to under 5 minutes**, eliminating timeout-driven retries.
- Integrated **AI-assisted development workflows** using **mcp servers** for code investigation, debugging, implementation assistance, and review workflows, reducing feature turnaround time by **30–40%** and accelerating issue resolution.

SDE 2

Aug 2023 - March 2025

- Drove **P95 latency** of the payroll dashboard APIs from **10s to 50ms** (~200×) via query aggregation optimisation, redis caching, and elimination of unnecessary round trips — removed the top customer-reported slowness tickets.
- Raised backend test coverage and reliability through **Mocha/Chai** unit + functional suites wired into CI gates; cut escaped defects in payroll modules by ~**38%** quarter-over-quarter.
- Stack:** NestJS, Node.js, TypeScript, MongoDB, Redis, Apache Kafka, BullMQ, Docker, Kubernetes, Jenkins, Mocha, Chai

Recrosoft Technologies — Synamedia (Cloud-Based Streaming Services)

Bangalore, India

Software Engineer

Dec 2020 - Aug 2023

- Built **BeIN Voice Support**, a voice-controlled Set-Top-Box experience integrating an **Alexa Video Skill** with backend Node.js microservices — delivered **sub-300ms** end-to-end directive response under production load.
- Implemented directive interpretation across all supported Alexa video interfaces (search, play, transport, channel control) — achieved **99.5% intent-completion accuracy** on the device fleet.
- Owned design and shipped a proxy notification service over an **MQTT broker**, replacing polling clients; cut message-delivery latency ~**75%** and lifted publish throughput ~**60%**.
- Profiled and tuned hot paths across the microservices (DB indexes, payload shape, connection reuse) — **30% faster** response times and **25% lower** server load at the same traffic.
- Improved observability across all the directive-processing (structured logs, traces, service-level metrics) — reduced production RCA time across skill interaction/notification incidents.
- Mentored 4 junior engineers** via code review, pair-programming sessions, and design walkthroughs, two were promoted within a year.
- Stack:** Node.js, TypeScript, Microservices, MongoDB, Redis, MQTT, Docker, Kubernetes, Jenkins

- Greenfielded **Time Dragon**, a B2B ERP unifying project management, employee data, client interactions, and timesheets — owned schema design, REST API layer, and admin dashboard end-to-end.
- Architected a real-time chat module on **Redis pub/sub + Socket.IO** with a horizontally-sharded socket registry — stress-tested to **1M concurrent connections** sustaining **2000 msg/sec**.
- Built **Jenkins**-based CI/CD pipelines (**Bash + Docker**) replacing manual deploys across staging and production — cut release lead time **~80%** and eliminated hand-rolled SSH steps.
- Shipped operational dashboard APIs using MongoDB aggregations to surface real-time user, project, and sales insights for **10k+ employees and 1200+ projects** across multiple business units, enabling data-driven decisions for leadership teams.
- **Stack:** Node.js, Express.js, TypeScript, MongoDB, PostgreSQL, Redis, Socket.IO, Jenkins, Docker

Education

Punjabi University, Patiala

2019

B.Tech in Computer Science

CGPA: 8.0/10

Relevant Coursework: Object Oriented Programming, DSA, DBMS, System Design

Achievements

- Betterplace 2025 Excellence Award (top performer)
- Recrosoft GEM Award
- Code-N-Counter Hackathon(Nagarro)